MAPS INTO A BOUNDED BELOW COMPLEX OF INJECTIVES ARE THE SAME IN K AND D

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 6769-6785, following the statement at 6757-6767.

Editorial note. No omissions: both directions delegate to fully carded lemmas.

Allowed standard prerequisites (editorial).

- Lifting against K-injectives. lemma-morphisms-lift (carded L26169, cited at ~6775). Surjectivity of Mor_K -> Mor_D.
- Parallel uniqueness. lemma-morphisms-equal-up-to-homotopy (carded L26190, cited at ~6783). Injectivity of Mor_K -> Mor_D.

Source reference. Original lemma 013S: sources/stacks-derived/pinned-original-source.tex, source lines 6698–6698. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 013P: sources/stacks-derived/pinned-original-source.tex, source lines 6603–6603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $I^\bullet$ be bounded below complex consisting of injective objects. Let $L^\bullet \in K(\mathcal{A})$. Then*

$$\mathrm{Mor}_{K(\mathcal{A})}(L^\bullet, I^\bullet) = \mathrm{Mor}_{D(\mathcal{A})}(L^\bullet, I^\bullet).$$

Proof. Let a be an element of the right hand side. We may represent $a = \gamma\alpha^{-1}$ where $\alpha : K^\bullet \to L^\bullet$ is a quasi-isomorphism and $\gamma : K^\bullet \to I^\bullet$ is a map of complexes. By Lemma 013P we can find a morphism $\beta : L^\bullet \to I^\bullet$ such that $\beta \circ \alpha$ is homotopic to γ. This proves that the map is surjective. Let b be an element of the left hand side which maps to zero in the right hand side. Then b is the homotopy class of a morphism $\beta : L^\bullet \to I^\bullet$ such that there exists a quasi-isomorphism $\alpha : K^\bullet \to L^\bullet$ with $\beta \circ \alpha$ homotopic to zero. Then Lemma 013S shows that β is homotopic to zero also, i.e., $b = 0$. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.